

Local resection versus combined local resection and plaque radiotherapy in the treatment of choroidal melanoma

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PURPOSE. *To evaluate the outcome of local resection versus combined local resection and plaque radiotherapy in the treatment of choroidal melanoma.*

METHODS. *We retrospectively analyzed the records of 60 patients (60 eyes) managed by local resection (group 1, 30 patients, external or internal local resection) or combined local resection and plaque radiotherapy (group 2, 30 patients, both Ruthenium-106 or Cobalt-60), among the 364 melanomas treated at our Department of Ophthalmology, between January 1980 and December 2006. Main outcomes measures were postsurgical complications, visual acuity, local recurrence, reasons for enucleation, and development of metastasis.*

RESULTS. *The 2 groups of patients were considered sufficiently homogeneous as regards location, cell type, and age. Mean follow-up was 7 years. The median largest basal tumor diameter was 9.8 mm and 10.1, and the median tumor thickness was 7.9 mm and 8, respectively, in groups 1 and 2. No statistically significant differences resulted between group 1 and group 2 as regards postsurgical complications (vitreous hemorrhage, retinal detachment, maculopathy, cataract, and optic neuropathy) ($p>0.05$), visual acuity ($p>0.05$), local recurrence ($p>0.05$), reasons for enucleation ($p>0.05$), or development of metastasis ($p>0.05$).*

CONCLUSIONS. *No adjunctive benefits from the combined local resection and plaque radiotherapy can be ruled out from our study. (Eur J Ophthalmol 2010; 20: 194-200)*

KEY WORDS. *Choroidal melanoma, Endoresection, Plaque radiotherapy, Transscleral resection*

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INTRODUCTION

Uveal melanomas are the most common primary intraocular malignancy and, other than the skin, the uvea is the site most commonly affected by melanoma (1). Choroidal melanoma has the potential to extend locally through the scleral wall and into the orbit and to metastasize to the liver and other extraocular sites (2). Death typically occurs within months after metastasis is detected clinically (2-4). The primary objective of treating the primary tumor is to reduce the threat of metastatic disease as much as possible (5). Enucleation has been the accepted treatment for primary choroidal melanoma for more than a century

(6), but the effectiveness of surgical removal of the eye to improve survival has not been unequivocally demonstrated because the natural history of uveal melanoma has never been documented (7, 8). The validity of these arguments was challenged by others, who believed that early enucleation offered the patient the best chance of cure (9-11). This controversy over enucleation has led to a trend away from enucleation and towards the increasing use of more conservative therapeutic methods. Such conservative methods include plaque radiotherapy, proton beam radiotherapy, stereotactic radiotherapy, photocoagulation, transpupillary thermotherapy, transscleral local resection, and transretinal endoresection (5, 12-19). The

2 most frequently employed treatment methods today are enucleation and episcleral plaque brachytherapy (1, 20). Local resection of posterior uveal melanoma offers several theoretical advantages over enucleation and radiotherapy (9, 21). Several reports have endorsed endoresection as a substitute for extraocular approaches in choroidal melanoma and for avoiding neuropathy associated with radiotherapy when the tumor is adjacent to the nerves (1, 19, 22-26). To our knowledge, there is no current evidence that local resection of posterior uveal melanoma is different from enucleation or radiotherapy with regard to patient survival (9). It appears that the rates of metastasis and mortality after enucleation, radiotherapy, and local resection of tumor are the same (1, 2, 8, 9, 22, 26-28). On the other hand, combining two or more conservative therapeutic methods seems to give better results (5, 9, 29).

The aim of our study is to compare the long-term outcomes of local resection versus combined local resection and plaque radiotherapy in the treatment of choroidal melanoma.

MATERIALS AND METHODS

In this retrospective study, we reviewed the charts of 364 patients with choroidal melanoma treated at our department between January 1980 and December 2006. Patients who underwent local resection were classified as group 1; patients who underwent combined local resection and plaque radiotherapy were classified as group 2. All procedures were performed by the same surgeon (N.D.). Patients were fully informed of the disease and its potential risks. All relevant aspects of the procedure, including the unusual nature of the associated treatments, were explained to the patients, and oral as well as written informed consent was obtained. Preoperative investigations included determination of best-corrected visual acuity, full ophthalmic examination, and A- and B-scan ultrasonography, with measurement of the maximal thickness and diameter of the tumor. Fundus photography, fluorescein angiography, magnetic resonance imaging of the eye and orbit, and screening for metastatic disease, which included serum biochemistry analysis with liver function tests, liver echography, and chest radiography, were also performed for each patient. Patients were excluded from the study if the tumor was bilateral, the tumor involved the iris, they had another cancer, or they had metastatic melanoma at the time of diagnosis. Eligible for transscleral local resection were choroidal mel-

anoma with pre-equatorial localization, largest basal diameter (LBD) <15.0 mm, and tumor thickness >6.0 mm. The inclusion criteria for endoresection were as follows: location of the posterior margin of the tumor >3 mm from the optic disk, absence of scleral invasion of the tumor, and anterior margin of the tumor not exceeding the equatorial area.

Postoperatively, the patients were generally discharged from the hospital after 2 days and then reviewed after approximately 1 week, 1 month, 12 weeks, and every 6 months thereafter for 3 years, and eventually once a year, indefinitely.

Main outcomes measures were postsurgical complications, visual acuity, local recurrence, reasons for enucleation, and development of metastasis.

For evaluation of distant metastases, the patients were also seen and examined by an internist or oncologist every 6 months and screened with at least liver function tests and chest x-rays.

Statistical calculations were performed using Epiinfo 6.05 software package (CDC, Atlanta). We performed univariate statistical analysis with chi-square test to assess the influence of adjunctive plaque brachytherapy to local resection on postsurgical outcomes. The chosen level of statistical significance was $p < 0.05$.

Surgical techniques

The surgical technique of external local resection was performed according to previously described techniques (18, 30). A 360° conjunctival peritomy was made. Extraocular muscles in the operative field were disinserted. Two traction sutures were placed in the sclera, approximately 4 mm from limbus. The tumor was localized by transpupillary transillumination, and the margins were marked on the sclera with a pen. A lamellar scleral flap was created, which was approximately 75% of the scleral thickness, hinged posteriorly, and with a polyhedral anterior margin. Any vortex veins in the operative field were cauterized outside the eye and within the sclera before being divided. The eye was decompressed by performing a limited three-port vitrectomy with a standard vitrector using illumination from the operating microscope and no infusion. When preparing the scleral trapdoor, deep scleral incisions were made approximately 1.5 mm within the superficial incisions, so as to create a stepped edge to facilitate closure. The tumor was excised together with the deep scleral lamella and a frill of normal uvea, if possible, without tearing the

retina or pars plana ciliary epithelium. Hemorrhage was controlled by systemic hypotensive anesthesia, lowering the systolic blood pressure to approximately 45 mmHg. The scleral window was closed with interrupted 8-0 nylon sutures. The muscles were reinserted with 6-0 braided polyglactin sutures, and the conjunctiva was closed with 7-0 braided polyglactin sutures. Injections of subconjunctival bupivacaine, gentamicin, and betamethasone were administered.

The technique of endoresection begins with complete three-port vitrectomy, with forced separation of the posterior vitreous and trimming of the vitreous base. A retinal incision is made in line with the nerve fiber bundle orientation with circumferential extension peripheral to the tumor to create a retinal flap. If, however, the tumor has invaded the retina, a direct approach is used. The retina over the tumor is detached, if necessary, by injecting fluid through the retinotomy, using a soft-tipped cannula. An incision is made in normal choroid around the tumor with vertically cutting scissors, if possible without damaging the overlying retina. Alternatively, choroidal vessels around the tumor are closed with diathermy. The ocular blood flow is reduced to a level at which the disc vessels are closed or almost closed, by lowering the systolic blood pressure to approximately 50–60 mmHg and if necessary by raising the infusion pressure. Before removing any instruments from the eye, the infusion pressure is lowered to avoid excessive retinal traction and tear formation near the sclerotomies. The tumor and surrounding normal choroids are removed piecemeal with the vitrector, leaving an area of bare sclera. After tumor removal, the infusion pressure is lowered until bleeding commences, so that all bleeding points can be recognized and treated by endodiathermy. The peripheral retina is examined, with indentation, so that if any entry site tears are found the overlying vitreous is removed and cryotherapy applied. Next, fluid-air exchange is performed so that all subretinal fluid drains into the choroidal coloboma, from where it is aspirated with a flute needle. After several minutes, once the retina has flattened, a double row of confluent endolaser burns is applied around the coloboma to create retinal adhesion. In addition, the entire bed of the coloboma and the margins of the choroid are treated with strong laser burns, to destroy any residual tumor cells. The eye is filled with silicone oil, which is removed after approximately 12 weeks. Cryotherapy is applied to the sclerotomies to destroy any tumor cells that may have seeded to these areas.

Cobalt-60 and ruthenium-106 brachytherapy was admin-

istered using custom-made, nonrimmed gold plaques (diameter 15, 20, and 25 mm; thickness 0.5 mm) after tumor resection (endo or transscleral). Before radioactive plaque application, any rectus muscle overlying the tumor is disinserted after placing two 6-0 Vicryl sutures in the muscle tendon and measuring the knot-to-limbus distances. The inferior oblique muscle is disinserted if it impeded contact of the plaque with the sclera and was held in its anatomic position by a sling. Tumors extending posteatorially are treated by placing the posterior edge of the plaque over the posterior margin of the resected area. A transparent plastic template is first sutured to the sclera with bows, and correct positioning is confirmed by indenting the eye with the posterior margin of the template while performing binocular indirect ophthalmoscopy. The template is then replaced with the radioactive plaque, using the same sutures. A mattress suture is placed over the plaque to ensure that it is firmly applied to the globe. Any disinserted muscles are returned to their correct anatomic position by slings attached to the sclera, the lugs of the plaque, or the mattress suture. The conjunctiva is closed with 7-0 braided polyglactin sutures. The prescription dose was usually 100 Gy to the apex of the tumor, allowing 1 mm for the thickness of the sclera.

RESULTS

A total of 60 eyes, of 60 examined patients, were included in the study and were assigned to 2 groups, based on surgical technique they underwent. Given the retrospective nature of this study, patients included for analysis were not randomized to the treatment; therefore, patients' assignment to 1 of the 2 groups simply reflected a retrospective selection of cases (showing homogeneous baseline characteristics) who underwent local resection or combined local resection and plaque radiotherapy. Group 1 patients underwent local resection: 25 patients underwent transscleral resection (Fig. 1) and 5 patients underwent endoresection (Fig. 2). Group 2 patients underwent combined local resection (25 patients underwent transscleral resection and 5 patients underwent endoresection) and plaque radiotherapy (22 patients underwent ruthenium-106 plaque radiotherapy and 8 patients underwent cobalt-60 plaque radiotherapy).

The 2 groups of patients were comparable with respect to location, cell type, sex, and age ($p>0.05$).

The mean age was 59 years in group 1 and 60 years in

Fig. 1 - Transscleral local resection. Color fundus photography pretreatment (A) and 1 year after transscleral local resection (B).

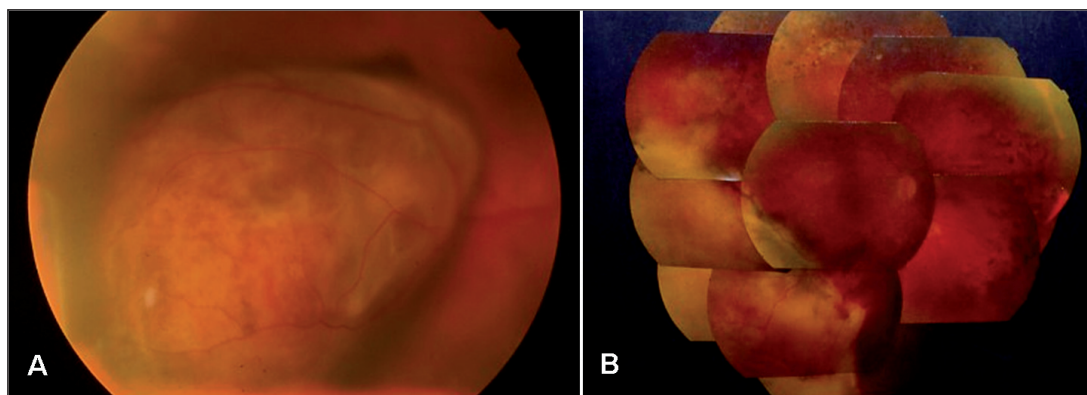


Fig. 2 - Endoresection. Color fundus photography pretreatment (A) and 1 year after endoresection (B).

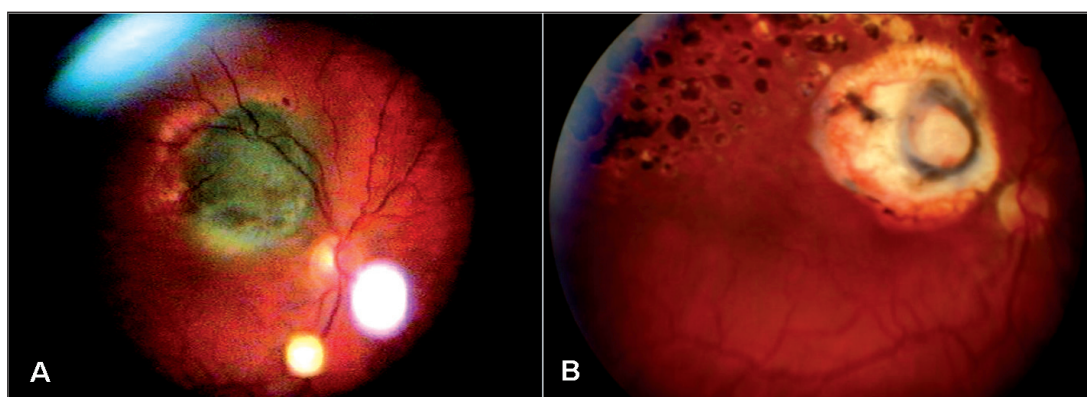
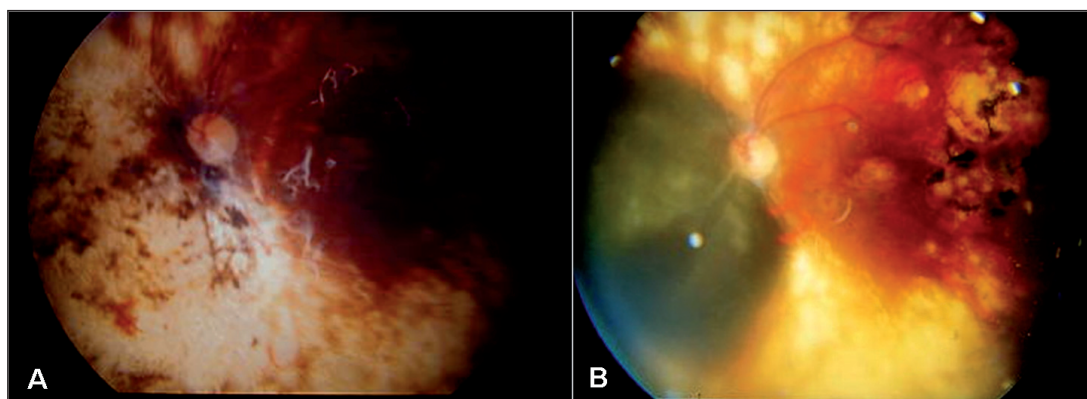


Fig. 3 - Color fundus photography 1 year after transscleral local resection (A) and recurrence 5 years after transscleral local resection (B).



group 2. The median follow-up time was 10.3 years (range 7–15 years). In group 1, the baseline median tumor thickness and LBD were 7.9 mm (range, 5.4–9.2 mm) and 9.8 mm (range, 6.7–12.9 mm), respectively. In group 2, the baseline median tumor thickness and LBD were 8.0 mm (range, 5.3–9.4 mm) and 10.1 mm (range, 6.9–13.7 mm), respectively. No statistically significant differences resulted between group 1 and group 2, as regards tumor thickness and LBD ($p>0.05$).

At baseline, vitreous hemorrhage was present in 1 eye (3.3%) in group 1 patients and in 0 eyes (0%) in group 2

patients; exudative retinal detachment was present in 20 eyes (66.7%) in group 1 patients and in 21 eyes (70%) in group 2 patients. Before treatment, only 5 patients (16.7%) in group 1 and 4 patients (13.3%) in group 2 had 1.0 vision, whereas 12 (40%) in group 1 and 13 (43.3%) in group 2 had a visual acuity of 0.1 or worse.

At 1 year follow-up, vitreous hemorrhage was present in 2 patients (6.7%) in either group 1 or group 2; retinal detachment was present in 3 patients (10%) in either group 1 or group 2; maculopathy was present in 7 patients (23.3%) in group 1 and in 8 patients (26.7%) in group 2; cataract

was present in 14 patients (46.7%) in group 1 and in 15 patients (50%) in group 2; optic neuropathy was present in 2 patients (6.7%) in group 1 and in 1 patient (3.3%) in group 2; bulbi phthisis was present in 0 patients in group 1 and in 1 patient (3.3%) in group 2; 15 patients (50 %) had visual acuity of 0.1 or better in group 1 and 14 (46.7 %) patients had visual acuity of 0.1 or better in group 2; local recurrence was present in 5 patients (16.7%) in group 1 and in 2 patients (6.7%) in group 2; there was no enucleation for bulbi phthisis in group 1 or group 2; 2 patients (6.7%) underwent enucleation for recurrence in group 1 and 1 patient (3.3%) underwent enucleation for recurrence in group 2; metastasis were present in 1 patient (3.3%) in either group 1 or group 2.

At 5 years follow-up, vitreous hemorrhage was present in 2 patients (6.7%) in group 1 and in 3 patients (10%) in group 2; retinal detachment was present in 3 patients (10%) in group 1 and in 4 patients (13.3%) in group 2; maculopathy was present in 8 patients (26.7%) in either group 1 or group 2; cataract was present in 16 patients (53.3%) in group 1 and in 18 patients (60%) in group 2; optic neuropathy was present in 2 patients (6.7%) in either group 1 or group 2; bulbi phthisis was present in 2 patients (6.7%) in either group 1 or group 2; 12 patients (40%) had visual acuity of 0.1 or better in group 1 and 13 (43.3%) patients had visual acuity of 0.1 or better in group 2; local recurrence was present in 9 patients (30%) in group 1 (Fig. 3) and in 3 patients (10%) in group 2; 1 patient (3.3%) underwent enucleation for bulbi phthisis in either group 1 or group 2; 5 patients (16.7%) underwent enucleation for recurrence in group 1 and 2 patients (6.7%) underwent enucleation for recurrence in group 2; metastasis was present in 4 patients (13.3%) in group 1 and in 3 patients (10%) in group 2.

At 7 years follow-up, vitreous hemorrhage was present in 2 patients (6.7%) in group 1 and in 3 patients (10%) in group 2; retinal detachment was present in 3 patients (10%) in group 1 and in 4 patients (13.3%) in group 2; maculopathy was present in 9 patients (30%) in either group 1 or group 2; cataract was present in 21 patients (70%) in group 1 and in 23 patients (76.7%) in group 2; optic neuropathy was present in 3 patients (10%) in group 1 in 2 patients (6.7%) in group 2; bulbi phthisis was present in 2 patients (6.7%) in either group 1 or group 2; 10 patients (33.3%) had visual acuity of 0.1 or better in group 1 and 12 (40%) patients had visual acuity of 0.1 or better in group 2; local recurrence was present in 11 patients (36.7%) in group 1 and in 5 patients (11.7%) in group 2; 2 patients (6.7%) underwent enucleation for bulbi phthisis in group 1 and 1 patient

(3.3%) underwent enucleation for bulbi phthisis in group 2; 7 patients (23.3%) underwent enucleation for recurrence in group 1 and 3 patients (10%) underwent enucleation for recurrence in group 2; metastasis was present in 8 patients (26.7%) in group 1 and in 7 patients (23.3%) in group 2.

No statistically significant differences resulted between group 1 and group 2, as regards postsurgical complications (vitreous hemorrhage, retinal detachment, maculopathy, cataract, and optic neuropathy) ($p>0.05$), visual acuity ($p>0.05$), local recurrence ($p>0.05$), reasons for enucleation ($p>0.05$), or development of metastasis ($p>0.05$).

DISCUSSION

Accuracy in diagnosing choroidal melanomas has improved in recent years as a result of more experience, a better understanding of risk factors, and improved diagnostic equipment such as ultrasound. Therefore, it has become much easier to identify cancerous lesions and begin treatment before metastasis occurs. In fact, detection of a small tumor has a more favorable prognosis than detection of a medium or large tumor, as evidenced by estimated 5-year mortality rates following enucleation of 16%, 32%, and 53% for small, medium, and large tumors, respectively (31). Furthermore, small tumor detection allows conservative treatments.

The optimal management of choroidal melanomas has long been a matter of controversy and the indications for various treatment modalities are continuously being refined and modified (1, 32, 33). Enucleation was historically the mainstay of treatments. However, because of the devastating consequences of removing an eye that still may be capable of useful vision, more effective and more conservative therapies to diagnose and treat the tumors are being investigated (31). Furthermore, in 1979, Zimmerman and colleagues (27, 34) proposed that enucleation of a globe with choroidal melanoma may actually decrease survival, noting the rise in the mortality rate in the years following enucleation from a baseline of 1% per year to a peak of 8% per year during the second year following enucleation. The rate subsequently returned to its baseline level of 1% per year over the next 3 to 5 years. The authors suggested 2 possible explanations: overwhelming tumor dissemination during surgery and a lowering of the host's immunologic defense (31).

It would appear that enucleation, radiotherapy, and even local resection have similar rates of metastases and death

(1, 35-37). We supposed that combined conservative treatments could have lower rates of metastases, local recurrence, and secondary enucleation. The purpose of this study, in fact, was to evaluate the differences, as regards postsurgical complications, visual acuity, local recurrence, reasons for enucleation, and development of metastasis, between local resection and combined local resection and plaque brachytherapy. Combining both treatments has several advantages. It enabled us to treat patients with tumors thicker than 5 mm, generally the maximum height for ruthenium-106 brachytherapy. The combination might also lead to a decreased radiation dose of the isotope used in brachytherapy (29, 38, 39). Local resection of posterior uveal melanoma offers several theoretical advantages over enucleation and radiotherapy. In contrast to enucleation, local resection of posterior uveal melanoma is designed to preserve vision and maintain a cosmetically acceptable eye. In contrast to radiotherapy, it has fewer long-term complications such as cataract or radiation retinopathy (9, 21). There are less complications and better visual results for smaller, more anteriorly located tumors than for larger postequatorial tumors (9, 21). The major disadvantage of this method is cutting into a malignant tumour and invariably liberating the tumor cells within the globe, some of which may grow after resection, leading to local tumor recurrence (9, 21).

In the current study, 21 eyes (70%) were saved and 9 eyes (30%) were enucleated in patients who underwent tumor local resection, while in patients who underwent combined local resection and plaque brachytherapy, 26 eyes (86.7%) were saved and 4 eyes (13.3%) were enucleated. Enucleation was due to local recurrence of tumor or to development of phthisis bulbi. Despite a difference in tumor recur-

rence rate between the 2 groups suggesting a lower rate in group 2 (at 5 years, 9 patients [30%] showed recurrence in group 1, and 3 patients [10%] in group 2), this difference did not reach significance ($p > 0.05$), probably because of the small sample size, and this represents the mean weakness of our study. Another weakness of this investigation is the way in which treatment evolved during the study period, since we used cobalt-60 plaque radiotherapy at the beginning of the study and ruthenium-106 plaque radiotherapy starting from the 1990s. Moreover, no specific definite criteria were identified and strictly followed on whether to add plaque radiotherapy to local resection. Finally, our study is likely to be flawed due to the retrospective nature: patients were not prospectively randomized to the treatment and assignment to 1 of the 2 groups simply reflected a retrospective selection of cases who underwent local resection or combined local resection and plaque radiotherapy. In conclusion, no adjunctive safety or efficacy from the combined local resection and plaque radiotherapy, when compared to local resection, can be ruled out from our study, even though further studies are needed to confirm these results.

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